Atharv Kiran Nawale

Pune, Maharashtra, India +91 9022618687 atharvnawale100@gmail.com Atharv428 atharvnawale04

Summary

Information Technology engineering student focused on AI/ML, deep learning, computer vision, and full-stack development. Experienced in developing Python-based semantic segmentation systems using PyTorch, OpenCV, NumPy, and Albumentations. Skilled in model training, data augmentation, and computer-vision evaluation using mIoU. Also experienced in building responsive web applications, REST APIs, and database-driven systems, with an interest in developing intelligent AI applications and agentic workflows.

Technical Skills

Data Analysis and Processing:	Pandas, NumPy, Matplotlib, Data Cleaning, Data Preprocessing , Image Segmentation,
AI/ML and Computer Vision:	PyTorch, OpenCV, DeepLabV3, YOLO , Object Detection, IoU-based Evaluation, mIoU
Programming Languages:	C, C++, Python, JavaScript
Frontend:	HTML, Tailwind CSS, React.js
Backend:	FastAPI, Node.js, Express.js
Databases:	SQL, MySQL, MongoDB, Supabase, Firebase
Cloud:	AWS, DigitalOcean
Tools:	Git, GitHub, VS Code, Linux (Ubuntu)
Core Concepts:	Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks

AI/ML Projects

Off-Road Semantic Segmentation GitHub
Technologies: *Python, PyTorch, OpenCV, NumPy, Albumentations, Matplotlib, Git, GitHub*

- Developed an AI-powered semantic segmentation system for pixel-level classification of off-road terrain images.
- Implemented and trained UNet and DeepLabV3 models with custom Dice Loss and Focal Loss using PyTorch.
- Applied Albumentations for data augmentation and evaluated model performance using Mean Intersection over Union (mIoU).
- Generated visual segmentation results for qualitative model evaluation.

Amazon ML Challenge 2026 – Business Entity Resolution GitHub
Technologies: *Python, Pandas, NumPy, LightGBM, RapidFuzz, Scikit-learn, Git, GitHub*

- Developed a machine learning pipeline to identify matching business entities across three heterogeneous data sources using business names, addresses, and country information.
- Implemented multi-key candidate generation and blocking to reduce the search space and enable scalable entity matching across millions of records.
- Engineered pairwise similarity features using RapidFuzz, including name and address similarity, token overlap, postal code matching, and country matching.
- Built a LightGBM classification pipeline with validation-based threshold calibration optimized for the Macro-$F_{0.5}$ evaluation metric.
- Implemented memory-efficient data processing, test inference, and TSV submission generation with support for previously unseen country labels.

Awards & Achievements

2nd Place – YOLO (You Only Launch Once) View Certificate
MIT-WPU, Pune

Amazon ML Challenge – 2026 View Certificate
Online-Unstop

Smart India Hackathon 2025 – Internal Round Qualified Twice View Certificate
NMIET, Pune

NPTEL Certification – Data Structures and Algorithms Design View Certificate
NPTEL (IIT Kanpur)

Code Arena 4.0 – DSA Coding Contest View Certificate
PCCOER Pune

GraphRAG Inference Hackathon by TigerGraph View Certificate

Education

B.E. in Information Technology 2024 – 2028
Nutan Maharashtra Institute of Engineering and Technology, Talegaon Dabhade, Maharashtra
Aggregate CGPA: 8.90/10
Relevant Coursework: Data Structures, Object-Oriented Programming, Database Management Systems, Operating Systems, Computer Networks